

Examining Inclusive Computing Education for Blind Students in India

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Abstract

The growing demand for computer professionals, driven by the expanding Information Technology industry, has led to numerous inclusive computing education efforts. These efforts have even included blind or visually-impaired (BVI) students, who are being increasingly encouraged to pursue education and a career in computing, despite the visually-oriented nature of the discipline. Extant literature has predominantly focused on identifying and addressing the accessibility barriers faced by BVI students to promote more inclusive learning environments. While few studies have also investigated the accessibility of computing education from the perspectives of BVI learners and instructors, these have been primarily situated in the Global North contexts; there is still a knowledge gap regarding the teaching and learning experiences of instructors and BVI students, respectively, in resource-constrained Global South contexts, where accessibility awareness and inclusion efforts are at nascent stages. To fill this gap, we conducted an interview study with 15 participants in India, where we inquired with BVI students, instructors, and BVI professionals, regarding their challenges, experiences, and needs pertaining to computing education. The study revealed that BVI students face significant difficulty in comprehending the instructional materials, the instructors often deal with courses not progressing as planned despite meticulous preparation, the students heavily depend on peer learning for grasping computing concepts, and they need additional support for managing the cognitively-burdensome task of simultaneously learning computing concepts and screen readers. Informed by the findings, we offer recommendations to improve computing curricula for BVI students and discuss self-learning assistive tools to supplement accessible computing education.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; **Accessibility technologies**; *User studies*.

Keywords

Blind, Visually impaired, Accessibility, Computing education

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1 Introduction

Ubiquitous digitization across industries has intensified global demand for computing talent. As a major hub for offshore Information and Communication Technology (ICT) services, India has expanded its computing education efforts to meet this need [35]. These efforts have included integration of computing fundamentals into school curricula (e.g., through the ‘National Education Policy 2020’ [26], the multi-lingual ‘AI for All’ program [27]) as well as expansion of digital education access for diverse demographics, particularly for underrepresented communities. Public-private collaborations, including platforms like FutureSkills Prime [30], Infosys Springboard [15], and IBM’s SkillsBuild [14], also play a key role in delivering targeted computing skill development to millions of learners across the country. Despite these initiatives, inclusive efforts to broaden participation of blind and visually impaired (BVI) individuals are still in their incipient stages, with organizations such as the National Association for the Blind (NAB) [31] and government initiatives like Digital India [28], Skill India [29], and Vision-Aid’s Project Springboard [40] starting to raise awareness in this regard [20].¹ Therefore, little is known about the experiences and challenges of BVI students and instructors in computing fields (in India); extant research in this regard has predominantly



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¹As of 2020, India is home to the largest population of people with visual impairments globally, including approximately 70 million individuals with vision impairments and 4.95 million who are blind [17, 43].

focused on the Global North developed countries where inclusive computing education is relatively mature (e.g., specialized training and workshops for teachers, accessibility in core curricula) and well-resourced (e.g., Offices of Educational Accessibility in universities [1]), compared to resource-constrained Global South developing countries such as India, where awareness of accessibility and inclusion is limited, even among computer science educators in K–12 and higher education.

To address this knowledge gap, we conducted an interview study with a diverse group of 15 BVI participants from India, which included BVI students (K12 and higher education), computer trainers, and working professionals, to answer these research questions:

- **RQ1:** What challenges do BVI students face in a developing society such as India regarding computing education?
- **RQ2:** What are the challenges and needs of instructors who teach computing to BVI students?

The study yielded unique insights pertaining to developing societies, notably, the instructors relied on a general curricula that focused on cultivating pre-defined software skills aimed at ensuring “job readiness”, rather than providing structured, hands-on computing instruction. Moreover, due to tight funding, most instructors were just volunteering proficient screen reader (SR) users rather than formally trained educators, which consequently limited their ability to explain computing concepts and address students’ queries and concerns. Informed by these findings, we finally discuss plausible solutions for enhancing inclusive computing education for BVI students in resource-limited developing countries and beyond.

2 Related Work

Inclusive computing education has increasingly focused on uncovering and removing barriers for students with disabilities, especially BVI students who encounter numerous challenges in traditional computer science courses, which often rely on visual content and tools (e.g. flowcharts and graphs). A seminal work by Mealin and Murphy-Hill [25] highlighted how blind software developers faced significant challenges with IDE accessibility and code navigation which they attributed to a lack of exposure and support for learning development tools. Baker et al. [2, 3] corroborated these findings and highlighted several factors contributing to the incomplete educational experiences of BVI students. These included inaccessible IDEs, non-engaging learning materials, limited teacher preparedness, increased reliance on peer support, the additional burden to master assistive technologies (ATs), and the need to self-teach through trial and error. Consequently, BVI students had less computing exposure compared to their sighted peers.

Building on these prior works, Huff et al. [13] found that teachers of BVI students were often forced to adapt or recreate lessons due to inaccessible course materials, faced a high cognitive burden when teaching development tools, and struggled with a lack of accessible and standardized curricula. They also observed that visual-centric tools and pedagogies often alienated blind students, while accessible alternatives like Quorum [38], though promising, still presented usability challenges.

To address these barriers, extant literature has explored a variety of alternative pedagogical approaches and intervention tools to make computing concepts more accessible to BVI students. For

instance, AccessCSforAll [21] is a National Science Foundation (NSF)-funded initiative that has adapted the AP Computer Science Principles curriculum to be fully accessible for BVI students by replacing visual programming tools with SR-friendly text-based alternatives like Quorum, and by training Teachers of the Visually Impaired (TVIs) to teach computing through audio and tactile methods. Stefik et al. [37] put this approach into practice by training TVIs to teach the AP Computer Science Principles course using the Quorum programming language, accessible screen-reader tools, and nonvisual teaching methods.

Bigham et al. [4] conducted an “Instant Messaging Chatbot” programming workshop for blind high school students, by using an accessible, dialogue-based coding scenario instead of the traditional GUI-centric one, which successfully inspired many participants to pursue Computer Science. In another workshop, Kane and Bigham [18] engaged blind students in programming by having them analyze live Twitter data during crisis events and generate 3D-printed models of the data, allowing students to physically explore their code’s output and deepen understanding through tactile feedback. A literature review by Hadwen-Bennett et al. [10] identified four key strategies for making programming accessible to BVI learners: (i) auditory and haptic feedback to convey visual information non-visually, (ii) adapted text-based programming languages designed with accessibility in mind, (iii) accessible block-based environments that support screen reader or keyboard navigation, and (iv) physical artifacts such as tangible programming tools. The review also noted a significant gap in the literature.

As evident, these efforts remain confined to affluent Global North contexts with relatively robust disability support and awareness, creating a notable knowledge gap regarding BVI computing education in resource-constrained Global South contexts. In these developing regions, accessibility adoption is still in its nascent stages, and instructors have fewer resources or training to support inclusive education [22]. Sparse literature concerning BVI individuals in developing countries, particularly India, predominantly focuses on general technology use, assistive technology adoption, voice-based social media engagement, and foundational digital literacy skills [9, 23, 24], with little attention given to their participation in structured computing education or the pedagogical challenges they face in such contexts.

Our work aims to fill this gap by triangulating perspectives from: (i) BVI students who navigate computing education firsthand, (ii) computer trainers and instructors who design and deliver instruction to BVI learners, and (iii) working BVI professionals who reflect on the efficacy of their educational experiences in shaping their careers in the ICT industry.

3 Method

We conducted semi-structured interviews with a diverse group of BVI participants from India, as described below.

3.1 Participants

We recruited 15 individuals² through mailing lists and word-of-mouth referrals from regional workforce development programs (training centers) in India. Selected participants met the following

²Studies with BVI users typically involve between 8 and 20 participants [13, 34].

ID	Age/ Gender	Age of Vision Loss	Occupation	Computer Experience (Years)	Prior Computer or Software training
P1	19/F	Since birth	College Student	2-3	Word, Notepad
P2	47/M	Don't Know	Computer Trainer	10-12	MS-Office, HTML & C Programming, VB script
P3	27/M	Since birth	Computer Trainer	8	MS-Office, Web Navigation, Call Center Training
P4	16/M	Age 7	Unemployed	less than 1	Basic Navigation, Notepad
P5	23/F	Age 10	College Student	5	MS-office, Data Entry, Web Navigation
P6	50/F	Sighted	Computer Teacher	18-20	Programming (C, JAVA, Python, C++)
P7	30/M	Don't Know	Unemployed	3-4	MS-Office, Web Navigation, Data Entry
P8	27/F	Age 15	Unemployed	6-7	Word, Notepad, Excel
P9	32/F	Don't Know	Bank Employee	7	MS-Office, Web Navigation
P10	41/F	Age 20/21	NGO Employee	5-7	Web Navigation, HTML, Notepad, Word
P11	39/M	Don't Know	Computer Trainer	11	MS-Office, Web Navigation
P12	20/F	Since Birth	Unemployed	2	Word, Notepad
P13	27/M	Since Birth	Graduate Student	9	HTML, MS-Office, Web Navigation, Latex
P14	47/M	Since birth	IT Employee	12-14	MS Office, Web Navigation, Python
P15	28/M	Sighted	Computer Teacher	10	Programming (JAVA, Python, C), Web Development

Table 1: Participant demographics. All information was self-reported by the participants.

inclusion criteria: (i) prior experience in a computing education program, and (ii) no hearing impairments that could affect their ability to participate in the interview study. Gender representation was approximately balanced, with 7 female and 8 male participants, ranging in age from 16 to 50 years.

The participant group included 5 teachers (3 BVI computer trainers and 2 computer teachers from regular K–12 schools), 7 BVI students currently enrolled or completed school (Unemployed), and 3 working professionals in computing fields. Table 1 provides self-reported demographic data.

3.2 Interview Design and Procedure

The questionnaire for the semi-structured interviews was developed based on established guidelines in qualitative research [32], and insights from prior studies involving BVI individuals [3, 19]. We designed separate sets of seed questions that aligned with our research questions for each participant group: students, teachers, and working professionals.

For BVI students, seed questions were related to learning, e.g., “What challenges did you face while learning computers and computing subjects?”, “How did you learn screen readers?”. Teachers were asked questions pertaining to instruction, e.g., “What strategies do you use to teach computing to BVI students?”, “What kind of training or resources have you received to support accessible instruction?”. For working professionals, questions elicited retrospection, e.g., “What do you wish you had learnt at school?”, “Given the computing industry demands, what should be included in the curriculum?”.

The study began with obtaining informed consent, followed by a basic demographic questionnaire to collect participant background information. Next up was the semi-structured interview which lasted for approximately an hour, allowing for in-depth discussions and rich descriptions of personal experiences. Inspired by Hove and Anda’s interview guidelines [11], we also asked follow-up clarification questions for responses that were unique, unclear, evasive, or

inconsistent with earlier answers. Each participant received a \$35.7 gift card (approximation after conversion from Indian Rupees) upon completion of the interview. All interviews were audio-recorded for data analysis.

3.3 Data Analysis

The annotators transcribed and if needed translated each of the transcripts into English to ensure uniformity in analysis. For analyzing transcribed data, we applied a hybrid thematic analysis method [5] that combined both inductive and deductive coding. First, each researcher inductively analyzed their respective transcripts using open coding [36]. This involved an iterative, line-by-line review of the interview transcripts to uncover themes and patterns in the data. These themes were grounded in the participants’ experiences, and in-vivo codes were used to preserve the interviewees’ exact wording, capturing the core of their stories.

Next, we proceeded to a deductive analysis where we mapped the codes derived from open coding onto established concepts from prior literature that aligned with our research questions. These concepts included self-learning, accessible education, technology adoption, and learning motivation. This process allowed us to examine how participants’ learning/teaching experiences either supported or challenged current knowledge. Upon completing both coding phases, all authors collaboratively reviewed, discussed, compared, and assimilated the codes and emergent themes from both the inductive and deductive approaches to reach a consensus.

4 Findings

The analysis revealed six notable themes described next.

4.1 Curriculum Structure

While most participants (11) (had) attended regular schools alongside sighted peers, following the same curriculum with exemptions

from visually-oriented “*practical exams*”, the remaining participants (had) only attended specialized schools for the blind, which followed a similar curriculum but included additional accessibility training in ATs. Regardless, all participants indicated that they (had) attended additional “*computer training centers*” (or *cram schools*) for honing their computer skills and knowledge.

The curricula of regular and specialized schools focused mainly on a theoretical approach to computer science aligned with foundational concepts, such as explanations of concepts like “*What is a disk?*” or “*What is the full form of USB?*”. The students and teachers typically relied on recorded lectures and, when available, Braille-based textbooks, using similar methods as they did for subjects like history or literature. BVI students were often granted exemptions from practical, hands-on computing activities, including exams that typically consisted only of theoretical questions, with no direct interaction with a computer (P13).

The computer training centers followed government recommended curriculum (KEONICS³) tailored to suit IT industry requirements (P3, P2, P11). While they offered practical hands-on training including coding, the curriculum predominantly focused on developing basic skills such as “*Data Entry or Call-center operator*,” which were often considered a “*respectful office job*” for blind individuals within their communities (P14, P9, P13).

In essence, the curriculum was largely application-specific, focusing heavily on MS Office, particularly Excel, with an emphasis on repetitive routine tasks commonly required in administrative or clerical roles (P14, P11). When asked about teaching ‘computing concepts’ such as IDEs and coding, the trainers (P2, P3, P11) indicated that these “*advanced concepts*” were taught only if students needed them for specific jobs. In such cases, trainers would teach skills like “*HTML coding or using the Command Prompt*”.

4.2 Instructional Challenges

All computer trainers (P2, P3, P11) at training centers stated that they did not receive formal training in pedagogical methods, and often faced difficulties addressing the diverse learning needs of BVI students. They further mentioned that they were just volunteers, i.e., the centers requested them to teach BVI students, due to their expertise in using screen readers and their proficiency in Office-related software applications and Web browsing. P11 explained,

I have studied BA in literature, then I joined as SR trainer here [his training center] to teach computer skills, as there is no money to hire professional teachers. I try to follow the given curriculum and teach what I know from my own experience. If I don't know something and a student wants to learn, I prepare and learn first, then teach them. For me also, no one is there to teach, I somehow manage it.

The trainers also mentioned that they struggled to customize instructions for students based on their specific visual conditions, age, technical capability, and linguistic backgrounds. For example, P2 mentioned that if any student struggled with a particular topic for some time, due to time constraints, he would simply instruct that student to engage in peer-learning with other students who

have understood the topic or those who speak the same language as the struggling student.

The computer teachers (P6, P15) in regular schools stated that they had received formal pedagogical training, but the training did not discuss much about accessibility and inclusiveness. They further mentioned that this lack of proper specialized training limited their ability to provide personalized support to BVI students. These teachers stated that their standard approach (as instructed by school management) to accommodating BVI students in their class was to simply make exemptions – skipping visually-oriented concepts, avoiding hands-on coding sessions, and theory-only exams without any practical computer tasks.

4.3 Peer-Guided Learning

The BVI students, as well as the working BVI professionals, stated that they (had) heavily relied on peer learning to grasp computing concepts, because the teachers were often unable to accommodate their individual learning needs. The BVI learners expressed that they often turned to their “*senior*” students who were more familiar with subjects and also proficient in efficiently performing computer tasks with a screen reader.

However, the students mentioned that such peer support was not always consistent. For example, P5 described:

“I would sometimes get help, but different people gave me different instructions, which was confusing. One would say, ‘Press Ctrl + S to save the file,’ while another would tell me to press ‘Alt, go to File, and then select Save.’ I was like: who should I listen to.”

Suggestion of different shortcuts to execute the same tasks caused BVI students to often second-guess themselves and question which approach was the “*right*” one (P1, P5). Additionally, the guidance was naturally limited to the knowledge of their peers, which often restricted their peer learning to routine application-specific tasks (e.g., applying a ribbon command in MS Word) without finding much help in programming activities or using developer tools (P7).

4.4 Self-learning Challenges

When asked about the extent of reliance on online learning sources for their computing education, all BVI students stated that, besides the instructor-provided subject material, they relied on either Google or YouTube to find manuals and tutorials pertaining to different subjects and concepts. However, most participants mentioned that these tutorials and ‘how-to’ videos were often inadequate, overly complex, or too generalized to offer context-specific solutions that they were seeking. P13 explained,

“The content is full of terms that someone new to this will not understand. It assumes you already know so much, which isn't the case for many of us... It either explains things in a way that doesn't make sense or it doesn't fully cover the exact issue I am dealing with.”

The participants also described difficulties in understanding the complex terminology used in online help manuals and tutorials. For instance, participants P4 and P8 noted they were initially unfamiliar with terms like “*ARIA*” or actions such as “*Toggle*”, which made it

³<https://www.keonics.in/english/courses>

challenging to determine how the explanation in the documentation applied to their specific issues.

Additionally, the participants sought a sequential list of actions (e.g., a sequence of keyboard shortcuts) as direct solutions to their problems, but such straightforward instructions were hardly available in the guides (P12). This issue extended to learning through online tutorial videos as well. P5 described his efforts to learn programming from YouTube videos, but found them difficult to follow:

“I tried to use YouTube tutorials, but it’s very hard to follow, even if they are accessible. The person gives instructions like click here or check the output, but he doesn’t describe what is happening. I need more detailed instructions, like what shortcut to press or what exactly to expect when. Also, sometimes they use terms I don’t understand, like terminal, path, or script. Normal people might know these easily, but for me it’s confusing without someone explaining slowly.”

When discussing the computer textbooks and lectures used in general education classes, which served as a primary guide for many students, P1 and P13 noted that these materials were largely focused on delivering general theoretical computer education. They seldom included guidance on executing real-world tasks using specific applications, interpreting auditory feedback, or navigating accessibility features. Additionally, the textbooks relied heavily on figures, which were often difficult for BVI students to interpret or relate to without informative alternative descriptions.

4.5 Need for Simultaneous Learning

All BVI participants identified a major challenge in their education: the simultaneous learning of both computing concepts and SR software. This dual learning process was not only time-intensive and cognitively demanding, but also posed a significant barrier before they could progress to any programming-related concepts.

Participant P9 described this dichotomy as, “it was not just about pressing keys” but rather involved “remembering where the keys are on the keyboard”, “decoding what the screen reader has said”, and “remembering how to respond” usually recalling from her past experiences. Participants P1, P7, and P13 had no prior exposure to computers or the structure of a QWERTY keyboard, including the location and meaning of keys and symbols, and required “a long time” to familiarize themselves with the keyboard.

The participants also described the complexity of scenarios where they needed to type in their native language (e.g., Hindi (P13)), which required them to not only learn the fundamentals of the SR and the keyboard but also navigate the intricate multilingual keyboard mappings between language characters, a challenge even for sighted users. P1 mentioned,

“Learning basic screen reader itself is challenging ... It took a long time for me to learn where each key is ... If I forget, then someone has to help. While navigating, I need to remember the key shortcut, find the key on the keyboard, then understand what the screen reader says. It takes so long to get used to it. My mind is always calculating, fixing errors, and it feels very slow. I get so tired and sometimes forget what I was learning in the first place.”

As a consequence, unlike their sighted peers, the participants spent a significant portion of their computer education time focused on learning SRs before progressing to learning actual computing concepts and subjects.

4.6 Employment after Education

Some of the BVI participants mentioned that although they graduated with a positive outlook and confidence in their computer skills post-education, they soon experienced professional setbacks. P14 noted, “I thought that after this education, I would be able to get a job. But once I started working, I realized that what we learned only scratched the surface.” While the education and additional center-provided training ensured basic computer skills and SR knowledge sufficient for tasks such as writing emails, coding algorithms, using MS Office, and developing websites, it did not prepare students for highly competitive industry-level digital tasks that required in-depth coding knowledge, familiarity with developer tools, and the ability to work independently without peer assistance. P14 highlighted this gap:

“I had to learn things like analyzing charts and using data science tools, but we were only taught MS Word and Excel at the [training] center. It feels like we are being left behind while everyone else is moving forward... Practically, why would they [software companies] hire someone blind with only the basics and spend money on our training and SR license?... We are competing with sighted people for jobs. So we should learn SR quickly and move to proper coding and development to do tasks on par with the normal population. I think that’s the only way... we need something to teach us SR early on and help us get used to ‘how to learn on our own’ different things, once that is done, we can learn by ourselves.”

After completing their education and additional training, some of the participants returned to their permanent residence (mostly in rural areas, with their family), as they were unable to support themselves for long periods due to the lack of immediate employment opportunities (P7, P8). In such circumstances, participants often lost access to computers, which they claimed led to a decline in their computer proficiency over time (P7, P8, P10).

5 Discussion

Our findings highlight significant gaps in computing education for BVI students in resource-constrained developing countries like India. First (RQ1), K–12 curricula for BVI students focus only on theoretical computer science with no practical components or accessibility support. Even the supplementary training centers tailor their instruction around routine Office-application tasks suited for secretarial or call center jobs.

Early K-12 training in accessibility, especially SRs, is absent, making it difficult later for students to focus on core computing concepts while simultaneously learning SRs. Individual learning needs are often unmet, online learning resources are less comprehensible, and peer learning is inconsistent and often confusing.

Second (RQ2), the teachers lack formal accessibility training, and BVI trainers are mostly volunteers with little knowledge of pedagogical preparation. The teachers also have limited time and

flexibility to adapt the curriculum for personalizing instruction to BVI students, which often results in the BVI students being left out of important computing subjects and even exams. To address these challenges, we next discuss plausible solutions that can potentially enhance computing education for BVI students in the face of resource constraints.

5.1 Computing curricula for Global South

Our findings reveal that, unlike computing education in the Global North, computing education in India reflects prevailing societal expectations that confine BVI students to predefined job roles such as transcription, clerical work, or call-center positions, rather than positioning computing as a viable career path. This is observable from the curriculum design, where only fundamental screen reader training is provided alongside computing concepts mostly in the later years of K-12 education, thereby erecting learning barriers for BVI students due to the reasons uncovered in our study.

We therefore recommend introducing SR learning much earlier in K-12 education, providing BVI students ample time to acquire computer-interaction proficiency before engaging with core computing subjects. While the cognitive demands of auditory interaction can be challenging during early developmental stages [39], curriculum design can draw on research showing that multi-sensory learning (i.e., combining auditory, tactile, and visual inputs) enhances cognitive development [7]. This approach can foster greater autonomy in computing education for BVI students and help them acquire general computer skills, allowing them to confidently delve into advanced computing concepts.

Besides the early introduction of SRs, we also recommend distributing the subject material across multiple K-12 grades, in contrast to the status quo, where all computing instruction is crammed into the last two K-12 grade levels. As the students depend more on self-learning and peer-learning due to the limited inclusive-instruction abilities of teachers and limited availability of other training resources, syllabus distribution across grades will provide BVI students the much-needed additional time to not only explore and comprehend the theoretical concepts, but also sufficiently practice and acquire practical computing skills such as coding.

5.2 ‘Smart’ Self-Learning Tools

While a complete overhaul of the curriculum and comprehensive training for computer trainers and instructors may take a long time to achieve in resource-constrained developing societies like India, we believe the current computing-education gaps can be partially bridged through affordable and accessible self-learning tools powered by AI techniques such as large language models [44]. Leveraging natural language processing, these tools can generate learning exercises, provide real-time feedback, and personalize instruction and assessment.

AI-driven tools are increasingly being integrated into K–12 and higher education settings in the Global North [6]. These tools can simulate human-like interactions and support young learners in exploring complex subjects, including mathematics and computational thinking [12, 42]. By expanding their scope to include BVI-specific learning support, these tools will be able to support a more inclusive computing education environment worldwide.

Comprehension issues in self-learning via online videos, tutorials, and documentation, as discussed under RQ1, can also be potentially addressed using Generative-AI tools, given their strong capabilities in language comprehension and translation tasks [41]. This opens up opportunities for multilingual learning assistants that offer real-time guidance to BVI learners in their preferred language – the kind of support that is often not available through conventional teaching methods. Such tools could reduce systemic costs and enhance the accessibility of computing education by eliminating the need for individuals to travel and live in another city to obtain computing education [8].

5.3 Limitations

Our participant pool consisted of BVI individuals from a single region of India, specifically the southern state of Karnataka, offering insights into the local challenges. However, India’s vast diversity in literacy, socioeconomic status, language, education systems, and urbanization [16, 33] suggests that training curricula and learning experiences may vary across the numerous states of India. Additional studies with diverse participant pools recruited from different parts of the country are therefore needed to assess the generalizability of our findings.

We included only participants with severe vision impairments, excluding those with moderate residual vision who may rely on a combination of assistive technologies, e.g., a screen magnifier together with a screen reader. The learning needs of people with moderate residual vision may significantly differ from those of people with severe blindness; therefore, separate studies are needed to uncover distinct learning needs and experiences pertaining to individuals belonging to the former group.

Our study relied solely on interview data, which may not fully capture ground realities. In future work, we plan to conduct in-person contextual inquiry studies across multiple computer training centers in India to observe and analyze in-situ instructional methods and teacher-student interactions.

6 Conclusion

This study examined inclusive computing education for BVI students in Global South contexts, focusing on India. Through semi-structured interviews with BVI students, computer trainers, and working professionals, the study uncovered context-specific challenges that hinder proficiency in computing among this marginalized group. Curricula were largely application-specific, teachers lacked the pedagogical training to support BVI learners, and peer knowledge was limited, which restricted progression beyond basic tasks. Simultaneous learning of screen readers and computing subjects was cognitively demanding. As a result, many BVI students abandoned computing careers and settled in clerical or customer service roles. The paper lastly discussed Global South-specific curricula modifications and smart self-learning tools that can potentially address these challenges and needs.

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